4304-Data Structures Lab. Winter 2022
Date: November 08, 2022.
Target Group: All
Topic: Trie

Instructions:

- Task naming format: fullID_T01L12_2A.c/CPP
- If you find any issues in the problem description/test cases, comment in the google classroom.
- If you find any tricky test case that I didn't include but others might forget to handle, please comment! I'll be happy to add.
- Use appropriate comments in your code. This will help you to recall the solution in the future easily.
- Obtained marks will vary based on the efficiency of the solution.
- Do not use <bits/stdc++.h> library, add the necessary libraries one after another.
- Do not use any built-in functions or any STL libraries. If the solution is extremely difficult without the usage of such components, discuss it with the instructors.
- Modified sections will be marked with BLUE color.

Group	**Tasks**
1B	1 2 3 4
2B	1 2 3 4
2A	1 2 3 4
1A	1 3 4 5

Task 1

Implement the basic operations of the 'Trie' data structure by implementing the following functions:

- void **Insert**() : Inserts a string in a trie
- boolean **Search**() : Returns if the query string is a valid word.
- void **Display**() : Shows all the words that are stored in the Trie in lexicographically sorted order.

The first line of input contains space-separated words that need to be inserted in the Trie. Once the words are inserted, display all of them.
The following line contains another collection of query words. Print **T/F** based on their presence/absence.

Sample Input	**Sample Output**
toy algo algorithm to tom also algea tommy toyota toy toyo al also algorithm algorithmic	algo algorithm algea also to tom tommy toy toyota T F F T T F

Task 2

Suppose a set of words is stored in a Dictionary. Given a *prefix*, your task is to find out how many words start with it.

The first input line will contain *N* and *Q*, where *N* represents the number of words in the dictionary, and Q is the number of queries. Print the number of words starting with each corresponding prefix.

Sample Input	**Sample Output**
10 10 Beauty Beast Beautiful Amazing Amsterdam Beautify Banana Xray Beauty Glorifying	
A	2
Am	2
AM	2
Beauty	2
Beaut	4
Beast	1
Ing	0
AMS	1
Be	5
B	6

Note: Convert every string/prefix in lowercase before storing/ searching.

Task 3

In English, we have a concept called root, which can be followed by some other word to form another longer word - let's call this word successor. For example, when the root "an" is followed by the successor word "other", we can form a new word "another".

Given a dictionary consisting of many roots and a sentence consisting of words separated by spaces, replace all the successors in the sentence with the root forming it. If a successor can be replaced by more than one root, replace it with the root that has the shortest length.

Return the sentence after the replacement.

Constraints:

- 1 <= dictionary.length <= 1000
- 1 <= dictionary[i].length <= 100
- dictionary[i] consists of only lower-case letters.
- 1 <= sentence.length <= 106
- sentence consists of only lower-case letters and spaces.
- The number of words in a sentence is in the range [1, 1000]
- The length of each word in the sentence is in the range [1, 1000]
- Every two consecutive words in a sentence will be separated by exactly one space.
- sentence does not have leading or trailing spaces.

Sample Input	**Sample Output**
cat bat rat the cattle was rattled by the battery	the cat was rat by the bat
a b c aadsfasf absbs bbab cadsfafs data base	a a b c data b
cat bat rat cut the cattle cuttle was rattled by the battery	the cat cut was rat by the bat

Task 4:

You are given an array of strings ‘products’ and a string ‘searchWord’.
Design a system that suggests at most **three** product names from products after each character of searchWord is typed. Suggested products should have a common prefix with searchWord. If there are more than three products with a common prefix return the three lexicographically minimums products.

Return a list of lists of the suggested products after each character of searchWord is typed. There will be length(searchWord) lines of suggestions.

Input	**Output**
mobile mouse moneypot monitor mousepad mouse	mobile moneypot monitor mobile moneypot monitor mouse mousepad mouse mousepad mouse mousepad
havana havana	havana havana havana havana havana
juice jeerapani icecream jelly jam jackfruit jalapeno jeans	jackfruit jalapeno jam Jelly jeerapani Null Null Null

Note:

- Explanation for Input 1:
  - Consider the products sorted lexicographically: mobile, moneypot, monitor, mouse, mousepad.
  - After typing ‘m’ and ‘mo’ all products match and we show moneypot, monitor.
  - After typing ‘mou’, ‘mous’ and ‘mouse’ the system suggests mouse, mousepad.
- Explanation for Input 2:
  - The only word ‘havana’ will be always suggested while typing the search word.

Task 5:

Given a list of phone numbers, determine if it is consistent in the sense that no number is the prefix of another. Let’s say the phone catalog listed the numbers:

- Emergency 911
- Alice 97625999
- Bob 91125426

In this case, it is not possible to call Bob, because the center will direct your call to the emergency line as soon as you had dialed the first three digits of Bob’s phone number. So this list is not considered to be consistent.

Given a list of phone numbers, your task is to print Yes/No based on whether the phone numbers are consistent or not.

Input	**Output**
911 97625999 91125426 -1	No
12340 123440 113 12345 98346 -1	Yes
12340 1234040 113 12345 98346 -1	No